

1 (a) State what is meant by a *progressive wave*.

9702/21/O/N/09/Q5

.....
..... [2]

(b) The variation with distance x along a progressive wave of a quantity y , at a particular time, is shown in Fig. 5.1.

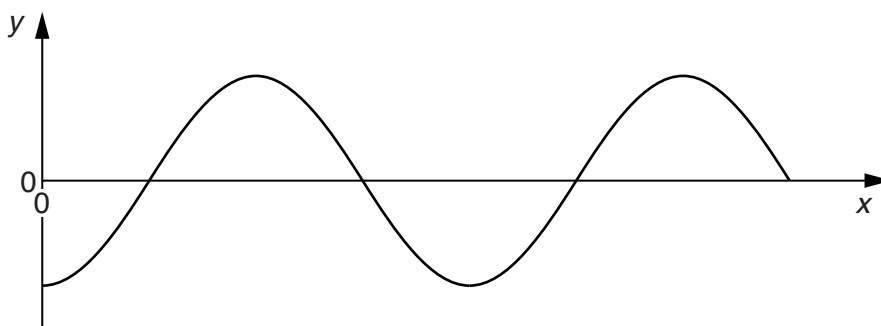


Fig. 5.1

(i) State what the quantity y could represent.

.....
..... [1]

(ii) Distinguish between the quantity y for

1. a transverse wave,

..... [1]

2. a longitudinal wave.

..... [1]

(c) The wave nature of light may be demonstrated using the phenomena of diffraction and interference.

Outline how diffraction and how interference may be demonstrated using light.

In each case, draw a fully labelled diagram of the apparatus that is used and describe what is observed.

diffraction

.....
.....

interference

.....
.....[6]

- 2 A uniform string is held between a fixed point P and a variable-frequency oscillator, as shown in Fig. 5.1.

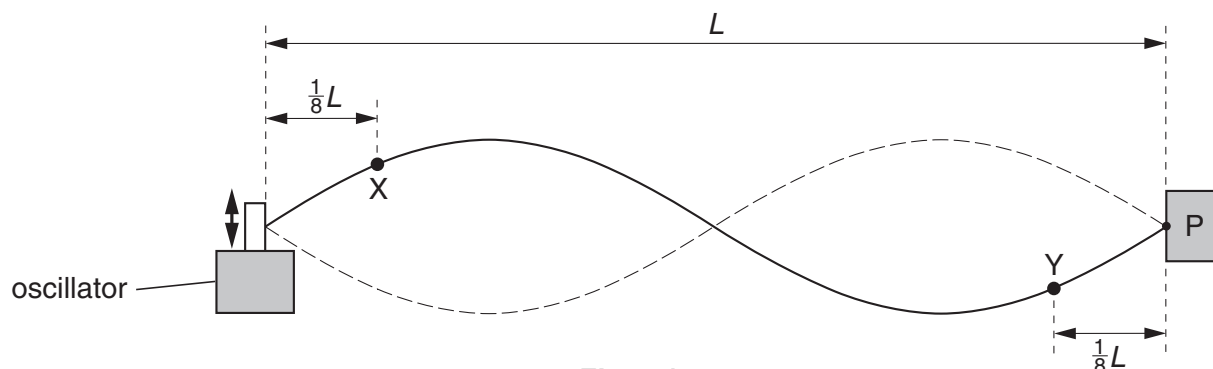


Fig. 5.1

The distance between point P and the oscillator is L .

The frequency of the oscillator is adjusted so that the stationary wave shown in Fig. 5.1 is formed.

Points X and Y are two points on the string.

Point X is a distance $\frac{1}{8}L$ from the end of the string attached to the oscillator. It vibrates with frequency f and amplitude A .

Point Y is a distance $\frac{1}{8}L$ from the end P of the string.

- (a) For the vibrations of point Y, state

- (i) the frequency (in terms of f),

frequency = [1]

- (ii) the amplitude (in terms of A).

amplitude = [1]

- (b) State the phase difference between the vibrations of point X and point Y.

phase difference = [1]

- (c) (i) State, in terms of f and L , the speed of the wave on the string.

speed = [1]

- (ii) The wave on the string is a stationary wave.

Explain, by reference to the formation of a stationary wave, what is meant by the speed stated in (i).

.....

 [3]

- 3 (a) State two features of a stationary wave that distinguish it from a progressive wave.

1.

.....

2.

..... [2]

- (b) A long tube is open at one end. It is closed at the other end by means of a piston that can be moved along the tube, as shown in Fig. 4.1.

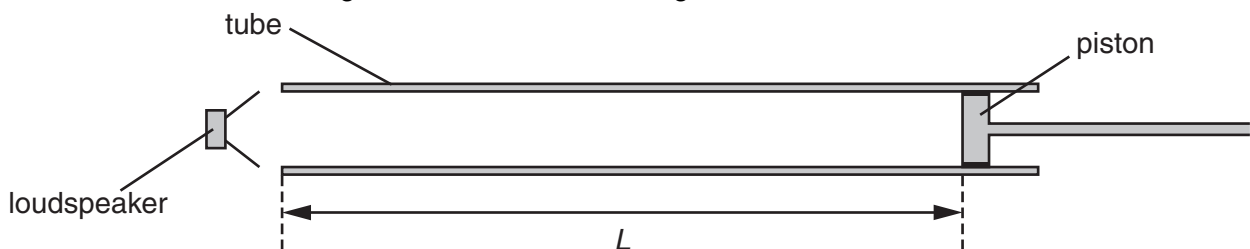


Fig. 4.1

A loudspeaker producing sound of frequency 550 Hz is held near the open end of the tube.

The piston is moved along the tube and a loud sound is heard when the distance L between the piston and the open end of the tube is 45 cm.

The speed of sound in the tube is 330 m s^{-1} .

- (i) Show that the wavelength of the sound in the tube is 60 cm.

[1]

- (ii) On Fig. 4.1, mark all the positions along the tube of

1. the displacement nodes (label these with the letter N),
2. the displacement antinodes (label these with the letter A).

[3]

- (c) The frequency of the sound produced by the loudspeaker in (b) is gradually reduced.

Determine the lowest frequency at which a loud sound will be produced in the tube of length $L = 45 \text{ cm}$.

frequency = Hz [3]

- 4 A student is studying a water wave in which all the wavefronts are parallel to one another. The variation with time t of the displacement x of a particular particle in the wave is shown in Fig. 5.1.

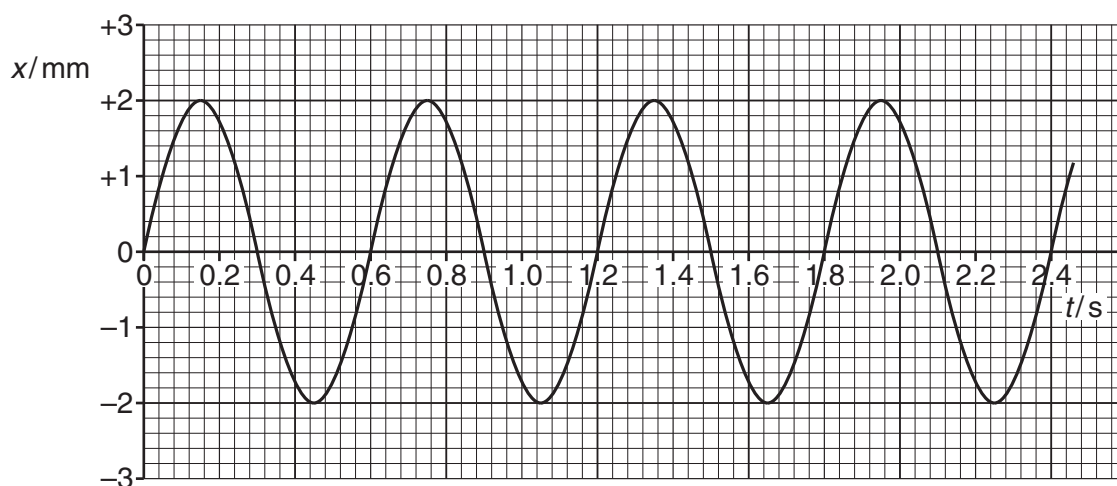


Fig. 5.1

The distance d of the oscillating particles from the source of the waves is measured. At a particular time, the variation of the displacement x with this distance d is shown in Fig. 5.2.

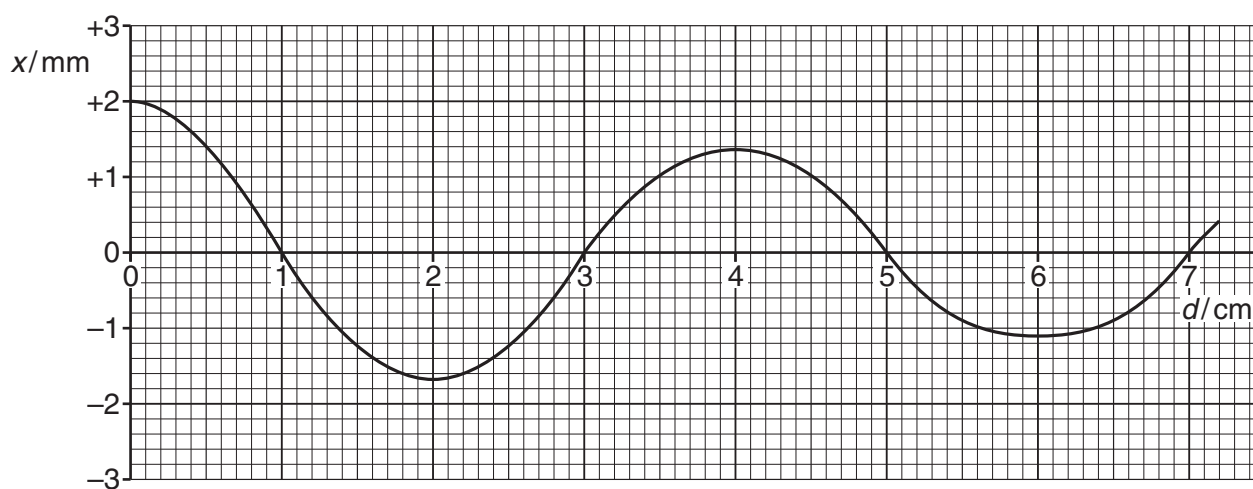


Fig. 5.2

- (a) Define, for a wave, what is meant by

(i) *displacement*,

.....[1]

(ii) *wavelength*.

.....[1]

(b) Use Figs. 5.1 and 5.2 to determine, for the water wave,

(i) the period T of vibration,

$T = \dots\dots\dots$ s [1]

(ii) the wavelength λ ,

$\lambda = \dots\dots\dots$ cm [1]

(iii) the speed v .

$v = \dots\dots\dots$ cm s⁻¹ [2]

(c) (i) Use Figs. 5.1 and 5.2 to state and explain whether the wave is losing power as it moves away from the source.

.....
.....
.....[2]

(ii) Determine the ratio

$$\frac{\text{intensity of wave at source}}{\text{intensity of wave 6.0 cm from source}} .$$

ratio = [3]

- 5 (a)** Use the principle of superposition to explain the formation of a stationary wave.

.....[3]

- (b)** Describe an experiment to determine the wavelength of sound in air using stationary waves. Include a diagram of the apparatus in your answer.

..... [3]

- (c) The variation with distance x of the intensity I of a stationary sound wave is shown in Fig. 6.1.

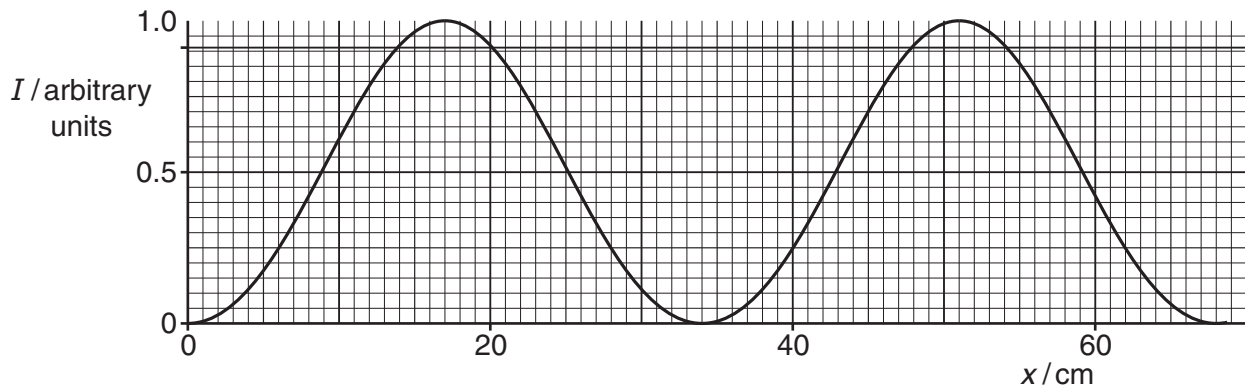


Fig. 6.1

- (i) On the x -axis of Fig. 6.1, indicate the positions of all the nodes and antinodes of the stationary wave. Label the nodes **N** and the antinodes **A**. [1]
- (ii) The speed of sound in air is 340 m s^{-1} .
Use Fig. 6.1 to determine the frequency of the sound wave.

frequency = Hz [3]

- 6 Fig. 4.1 shows an arrangement for producing stationary waves in a tube that is closed at one end.

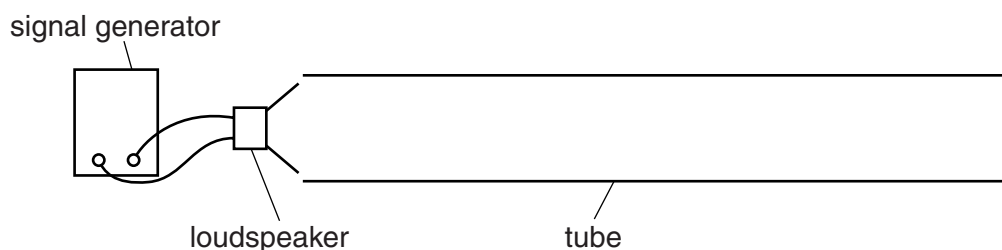


Fig. 4.1

- (a) Explain how waves from the loudspeaker produce stationary waves in the tube.

.....

.....

.....

.....

..... [3]

- (b) One of the stationary waves that may be formed in the tube is represented in Fig. 4.2.

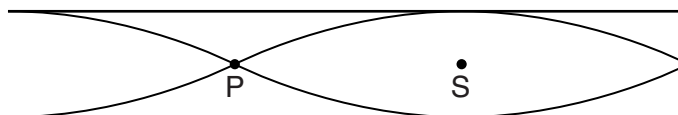


Fig. 4.2

- (i) Describe the motion of the air particles in the tube at

1. point P,

..... [1]

2. point S.

..... [1]

- (ii) The speed of sound in the tube is 330 ms^{-1} and the frequency of the waves from the loudspeaker is 880 Hz . Calculate the length of the tube.

length = m [3]

- 7 (a) State one property of electromagnetic waves that is **not** common to other transverse waves.

..... [1]

- (b) The seven regions of the electromagnetic spectrum are represented by blocks labelled A to G in Fig. 5.1.

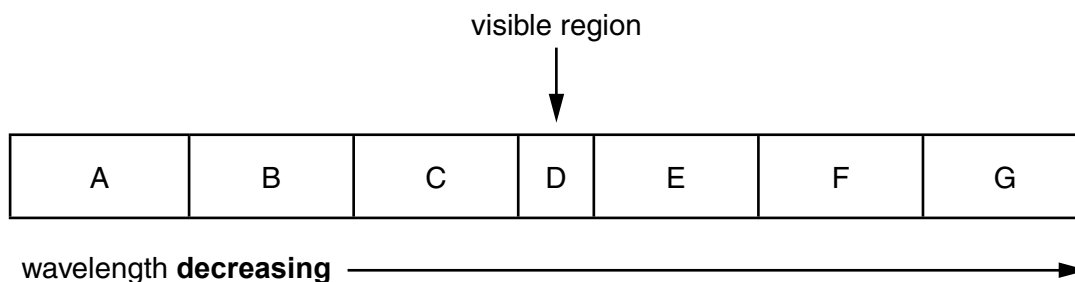


Fig. 5.1

A typical wavelength for the visible region D is 500 nm.

- (i) Name the principal radiations and give a typical wavelength for each of the regions B, E and F.

B: name: wavelength: m

E: name: wavelength: m

F: name: wavelength: m
[3]

- (ii) Calculate the frequency corresponding to a wavelength of 500 nm.

frequency = Hz [2]

- (c) All the waves in the spectrum shown in Fig. 5.1 can be polarised. Explain the meaning of the term *polarised*.

.....

.....

.....

..... [2]

8 Fig. 5.1 shows a string stretched between two fixed points P and Q.

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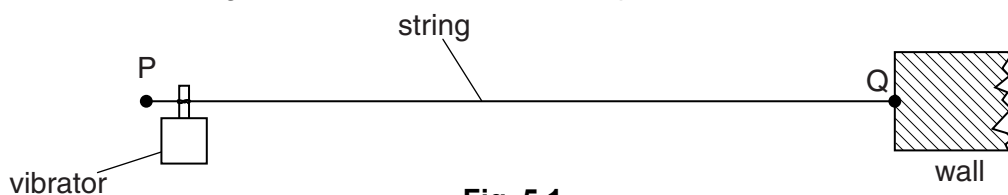


Fig. 5.1

A vibrator is attached near end P of the string. End Q is fixed to a wall. The vibrator has a frequency of 50 Hz and causes a transverse wave to travel along the string at a speed of 40 m s^{-1} .

(a) (i) Calculate the wavelength of the transverse wave on the string.

wavelength = m [2]

(ii) Explain how this arrangement may produce a stationary wave on the string.

.....

 [2]

(b) The stationary wave produced on PQ at one instant of time t is shown on Fig. 5.2. Each point on the string is at its maximum displacement.

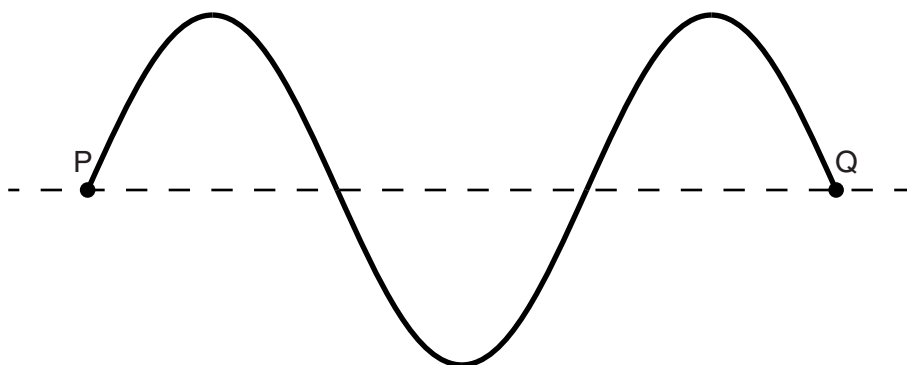


Fig. 5.2 (not to scale)

(i) On Fig. 5.2, label all the nodes with the letter **N** and all the antinodes with the letter **A**. [2]

(ii) Use your answer in (a)(i) to calculate the length of string PQ.

length = m [1]

(iii) On Fig. 5.2, draw the stationary wave at time $(t + 5.0 \text{ ms})$. Explain your answer.

..... [3]

- 9 A long rope is held under tension between two points A and B. Point A is made to vibrate vertically and a wave is sent down the rope towards B as shown in Fig. 5.1.

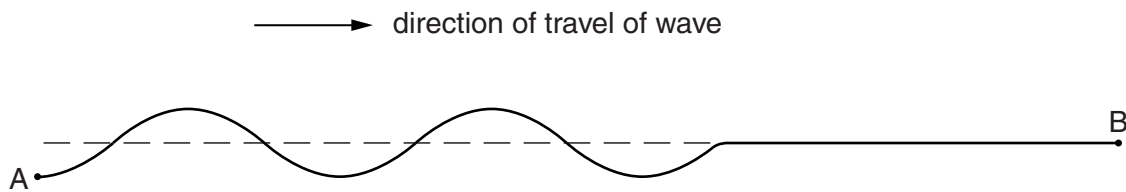


Fig. 5.1 (not to scale)

The time for one oscillation of point A on the rope is 0.20 s. The point A moves a distance of 80 mm during one oscillation. The wave on the rope has a wavelength of 1.5 m.

- (a) (i) Explain the term *displacement* for the wave on the rope.

.....
 [1]

- (ii) Calculate, for the wave on the rope,

1. the amplitude,

amplitude = mm [1]

2. the speed.

speed = m s^{-1} [3]

- (b) On Fig. 5.1, draw the wave pattern on the rope at a time 0.050 s later than that shown. [2]

- (c) State and explain whether the waves on the rope are

- (i) progressive or stationary,

.....
 [1]

- (ii) longitudinal or transverse.

.....
 [1]

- 10 (a) (i) Define, for a wave,
1. wavelength λ ,

..... [1]

2. frequency f .

..... [1]

- (ii) Use your definitions to deduce the relationship between λ , f and the speed v of the wave.

[1]

- (b) Plane waves on the surface of water are represented by Fig. 5.1 at one particular instant of time.

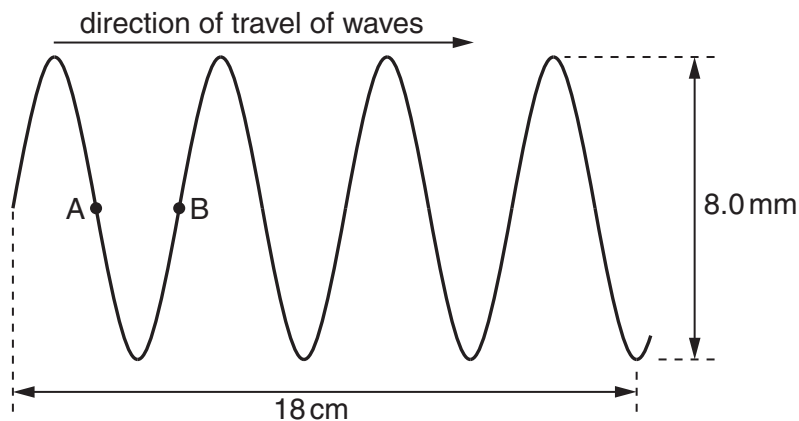


Fig. 5.1 (not to scale)

The waves have frequency 2.5 Hz.

Determine, for the waves,

- (i) the amplitude,

amplitude = mm [1]

- (ii) the speed,

speed = ms^{-1} [2]

- (iii) the phase difference between points A and B.

phase difference = unit [1]

- (c) The wave in (b) was produced in a ripple tank. Describe briefly, with the aid of a sketch diagram, how the wave may be observed.

.....

 [2]

11 (a) Explain what is meant by the following quantities for a wave on the surface of water:

(i) displacement and amplitude,

displacement

amplitude [2]

(ii) frequency and time period.

frequency

time period [2]

(b) Fig. 5.1 represents waves on the surface of water in a ripple tank at one particular instant of time.

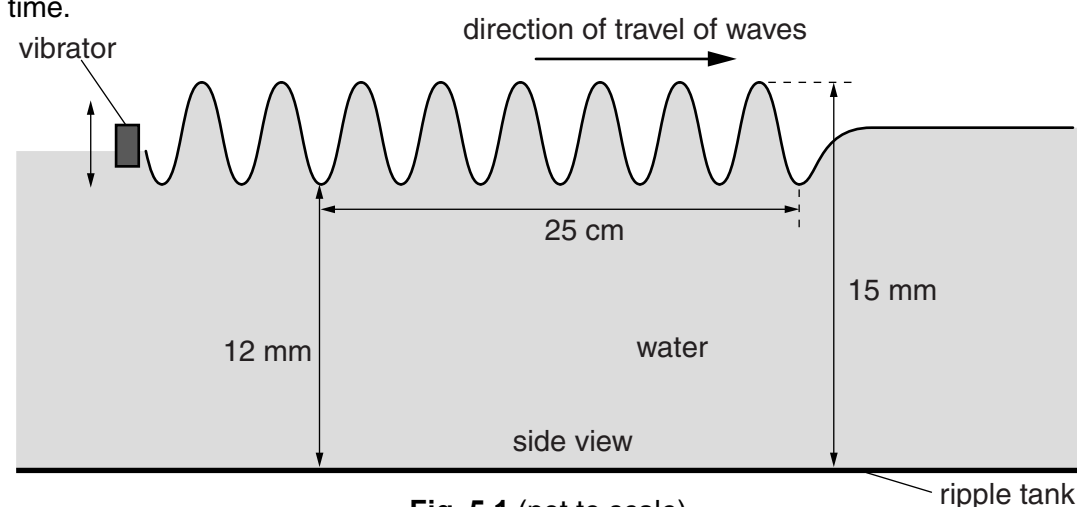


Fig. 5.1 (not to scale)

A vibrator moves the surface of the water to produce the waves of frequency f . The speed of the waves is 7.5 cm s^{-1} . Where the waves travel on the water surface, the maximum depth of the water is 15 mm and the minimum depth is 12 mm.

(i) Calculate, for the waves,

1. the amplitude,

amplitude = mm [1]

2. the wavelength.

wavelength = m [2]

(ii) Calculate the time period of the oscillations of the vibrator.

time period = s [2]

(c) State and explain whether the waves on the surface of the water shown in Fig. 5.1 are

(i) progressive or stationary,

..... [1]

(ii) transverse or longitudinal.

..... [1]

- 12 A hollow tube is used to investigate stationary waves. The tube is closed at one end and open at the other end. A loudspeaker connected to a signal generator is placed near the open end of the tube, as shown in Fig. 6.1.

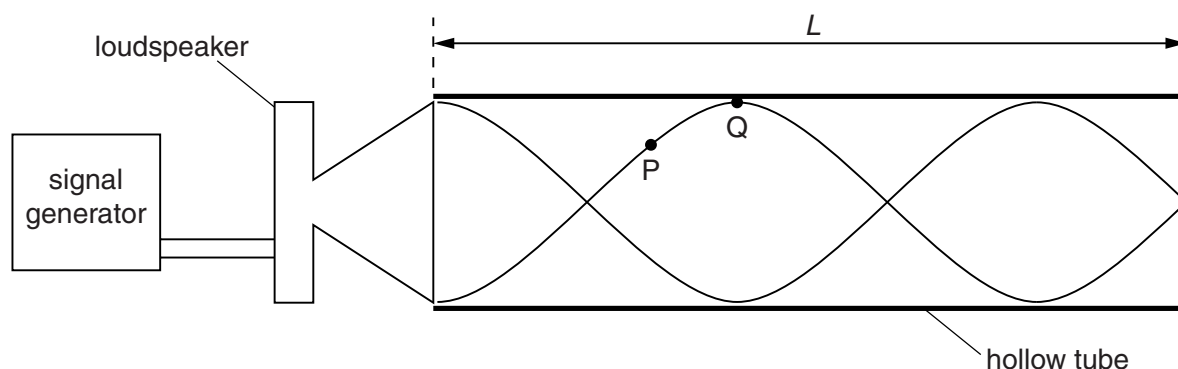


Fig. 6.1

The tube has length L . The frequency of the signal generator is adjusted so that the loudspeaker produces a progressive wave of frequency 440 Hz. A stationary wave is formed in the tube. A representation of this stationary wave is shown in Fig. 6.1.

Two points P and Q on the stationary wave are labelled.

- (a) (i) Describe, in terms of energy transfer, the difference between a progressive wave and a stationary wave.

.....[1]

- (ii) Explain how the stationary wave is formed in the tube.

.....

.....

.....[3]

- (iii) State the direction of the oscillations of an air particle at point P.

.....[1]

- (b) On Fig. 6.1 label, with the letter N, the nodes of the stationary wave. [1]

- (c) State the phase difference between points P and Q on the stationary wave.

phase difference = [1]

- (d) The speed of sound in the tube is 330 m s^{-1} .

Calculate

- (i) the wavelength of the sound wave,

wavelength = m [2]

- (ii) the length L of the tube.

length = m [2]

13 (a) Explain how stationary waves are formed.

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.....
.....
..... [2]

(b) The arrangement of apparatus used to determine the wavelength of a sound wave is shown in Fig. 8.1.

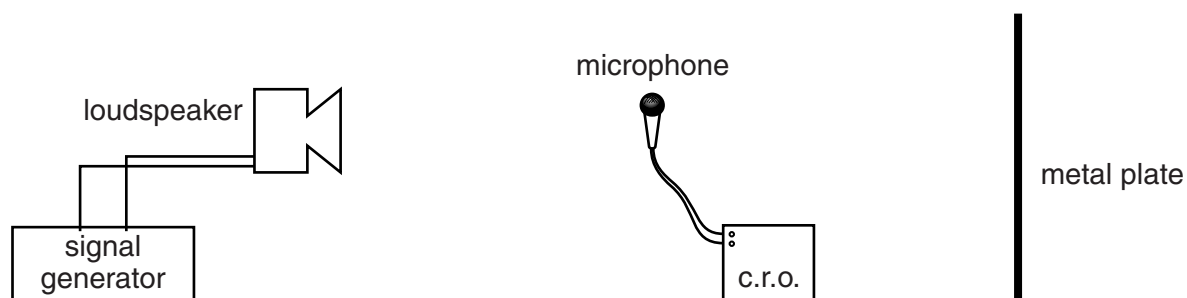


Fig. 8.1

The loudspeaker emits sound of one frequency. The microphone is connected to a cathode-ray oscilloscope (c.r.o.).

The waveform obtained on the c.r.o. for one position of the microphone is shown in Fig. 8.2.

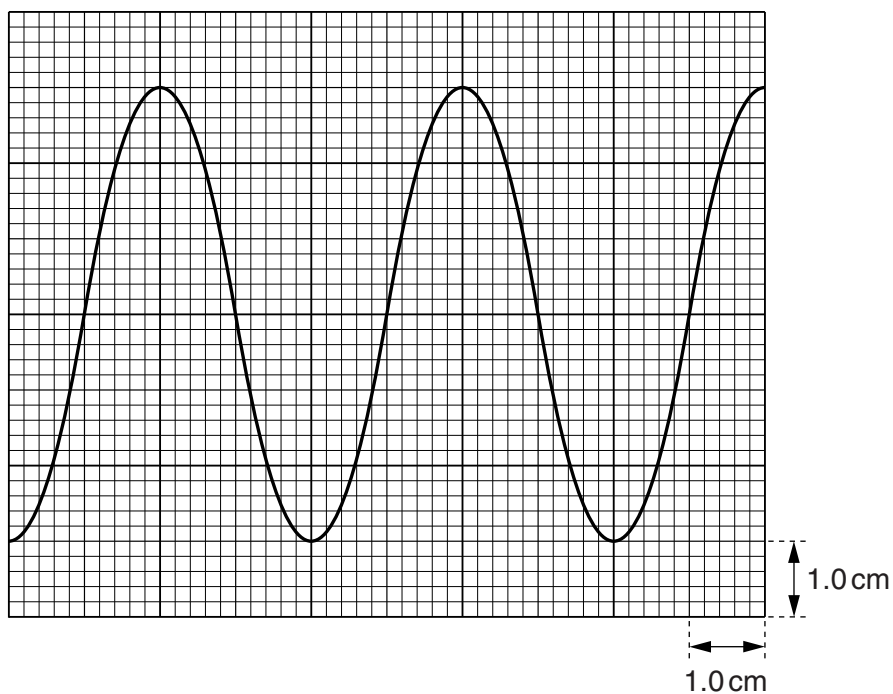


Fig. 8.2

The time-base setting of the c.r.o. is 0.20 ms cm^{-1} .

(i) Use Fig. 8.2 to show that the frequency of the sound is approximately 1300 Hz.

(ii) Explain how the apparatus is used to determine the wavelength of the sound. [2]

.....
.....
.....
..... [2]

(iii) The wavelength of the sound wave is 0.26 m. Calculate the speed of sound in this experiment.

speed = ms^{-1} [2]

9702/22/O/N/14/Q1

14 (a) The Young modulus of the metal of a wire is $1.8 \times 10^{11} \text{ Pa}$. The wire is extended and the strain produced is 8.2×10^{-4} . Calculate the stress in GPa.

stress = GPa [2]

(b) An electromagnetic wave has frequency 12 THz.

(i) Calculate the wavelength in μm .

wavelength = μm [2]

(ii) State the name of the region of the electromagnetic spectrum for this frequency.

.....[1]

15 (a) (i) Explain what is meant by a *progressive transverse wave*.

9702/23/O/N/14/Q7

progressive:

.....

transverse:

.....

[2]

(ii) Define frequency.

.....

.....[1]

(b) The variation with distance x of displacement y for a transverse wave is shown in Fig. 7.1.

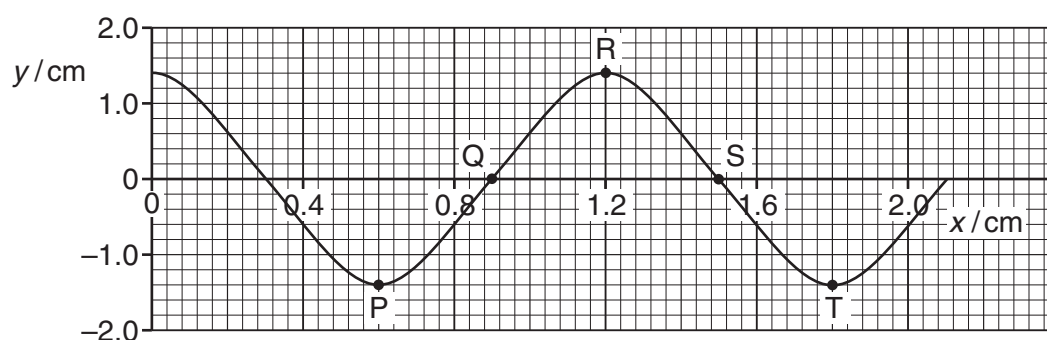


Fig. 7.1

On Fig. 7.1, five points are labelled.

Use Fig. 7.1 to state any two points having a phase difference of

(i) zero,

.....[1]

(ii) 270° .

.....[1]

(c) The frequency of the wave in (b) is 15 Hz.

Calculate the speed of the wave in (b).

speed = ms^{-1} [3]

(d) Two waves of the same frequency have amplitudes 1.4 cm and 2.1 cm.

Calculate the ratio

$$\frac{\text{intensity of wave of amplitude 1.4 cm}}{\text{intensity of wave of amplitude 2.1 cm}}$$

ratio = [2]

16 (a) State two differences between progressive waves and stationary waves. **9702/22/M/J/15/Q6**

1.

2.

..... [2]

(b) A source S of microwaves is placed in front of a metal reflector R, as shown in Fig. 6.1.

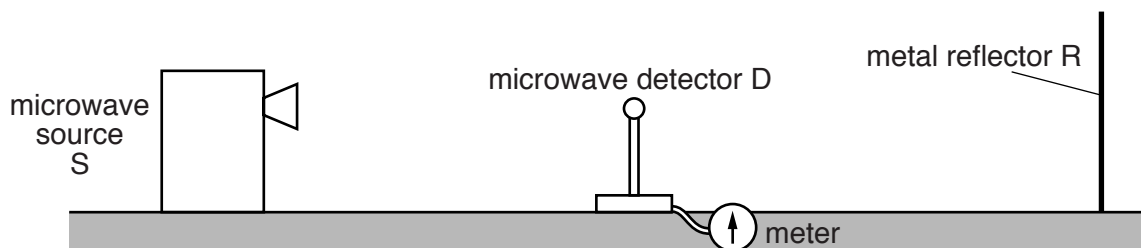


Fig. 6.1

A microwave detector D is placed between R and S.

Describe

(i) how stationary waves are formed between R and S,

.....

.....

..... [3]

(ii) how D is used to show that stationary waves are formed between R and S,

.....

..... [2]

(iii) how the wavelength of the microwaves may be determined using the apparatus in Fig. 6.1.

.....

..... [2]

(c) The wavelength of the microwaves in **(b)** is 2.8cm. Calculate the frequency, in GHz, of the microwaves.

frequency = GHz [3]

- 17 A wave is produced on the surface of a different liquid. At one particular time, the variation of the vertical displacement y with distance x along the surface of the liquid is shown in Fig. 4.2.

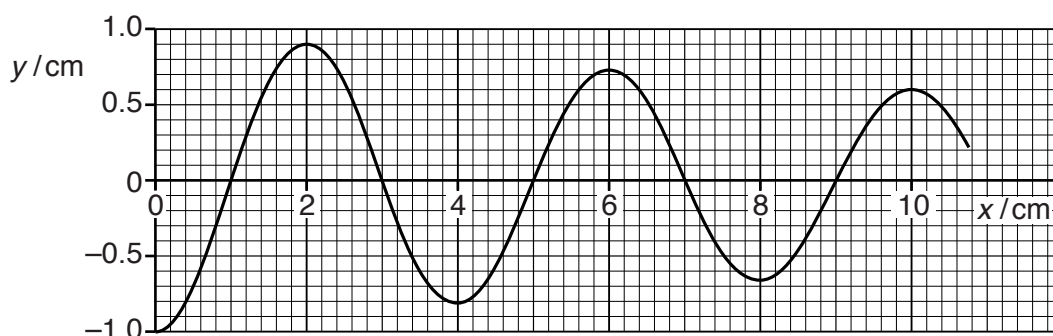


Fig. 4.2

- (i) The wave has intensity I_1 at distance $x = 2.0$ cm and intensity I_2 at $x = 10.0$ cm.

Determine the ratio

$$\frac{\text{intensity } I_2}{\text{intensity } I_1}.$$

ratio = [2]

- (ii) State the phase difference, with its unit, between the oscillations of the liquid particles at distances $x = 3.0$ cm and $x = 4.0$ cm.

phase difference = [1]

- 18 The variation with time t of the displacement y of a wave X, as it passes a point P, is shown in Fig. 5.1.

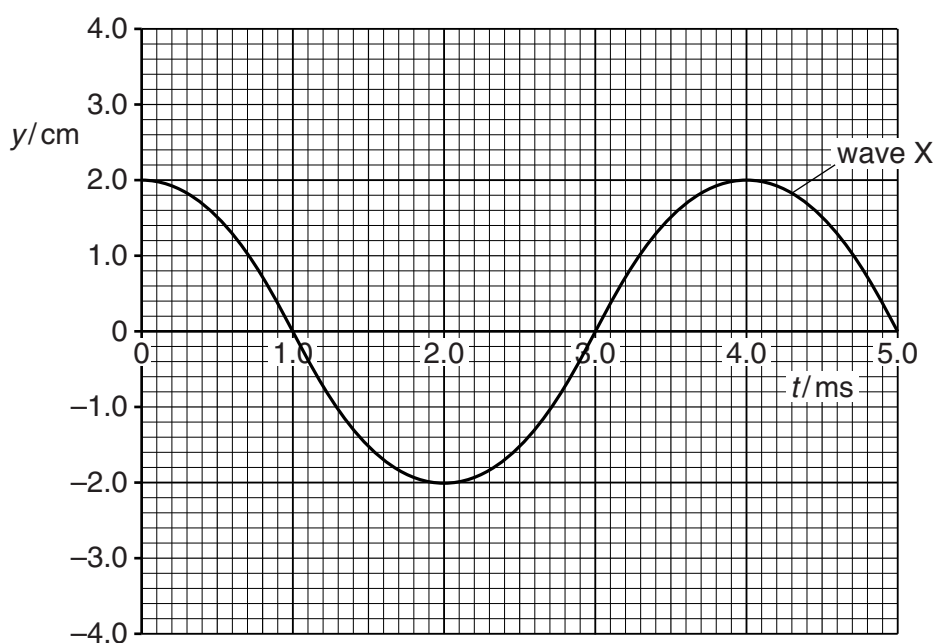


Fig. 5.1

The intensity of wave X is I .

- (a) Use Fig. 5.1 to determine the frequency of wave X.

frequency = Hz [2]

- (b) A second wave Z with the same frequency as wave X also passes point P. Wave Z has intensity $2I$. The phase difference between the two waves is 90° .

On Fig. 5.1, sketch the variation with time t of the displacement y of wave Z.

Show your working.

- 19 (a) By reference to the direction of the propagation of energy, state what is meant by a *longitudinal* wave and by a *transverse* wave.

longitudinal:

.....

.....

transverse:

.....

.....

[2]

- (b) The intensity of a sound wave passing through air is given by

$$I = K v \rho f^2 A^2$$

where I is the intensity (power per unit area),

K is a constant without units,

v is the speed of sound,

ρ is the density of air,

f is the frequency of the wave

and A is the amplitude of the wave.

Show that both sides of the equation have the same SI base units.

[3]

(c) (i) Describe the *Doppler effect*.

.....
.....[1]

(ii) A distant star is moving away from a stationary observer.

State the effect of the motion on the light observed from the star.

.....
.....
.....[1]

(d) A car travels at a constant speed towards a stationary observer. The horn of the car sounds at a frequency of 510 Hz and the observer hears a frequency of 550 Hz. The speed of sound in air is 340 m s^{-1} .

Calculate the speed of the car.

speed = m s^{-1} [3]

- 20 (a) Apparatus used to produce stationary waves on a stretched string is shown in Fig. 7.1.

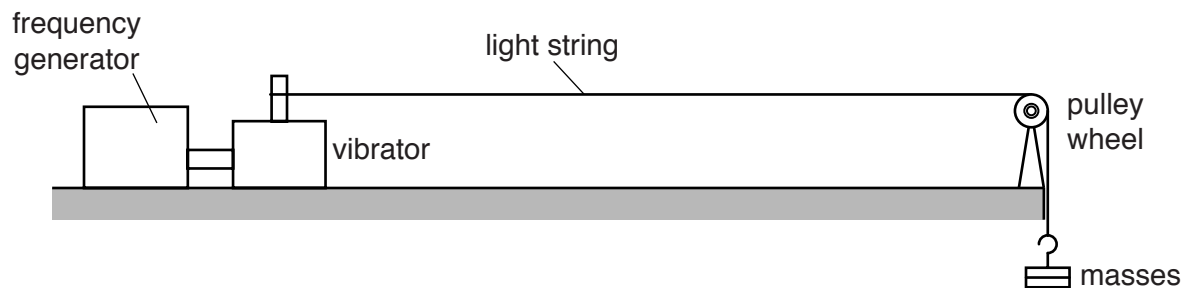


Fig. 7.1

The frequency generator is switched on.

- (i) Describe two adjustments that can be made to the apparatus to produce stationary waves on the string.

1.

.....

2.

.....

[2]

- (ii) Describe the features that are seen on the stretched string that indicate stationary waves have been produced.

..... [1]

- (b) The variation with time t of the displacement x of a particle caused by a progressive wave R is shown in Fig. 7.2. For the same particle, the variation with time t of the displacement x caused by a second wave S is also shown in Fig. 7.2.

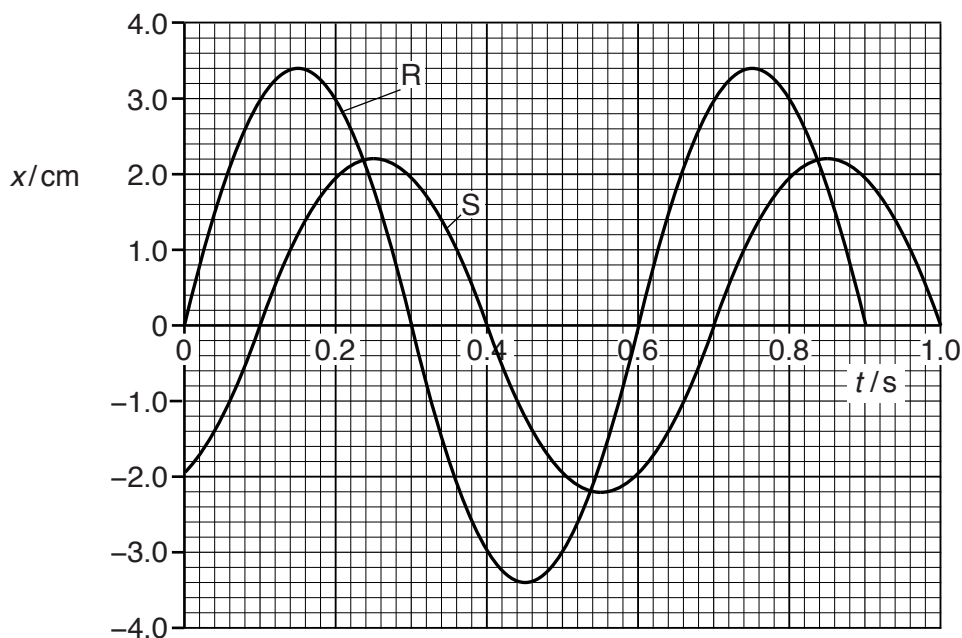


Fig. 7.2

- (i) Determine the phase difference between wave R and wave S. Include an appropriate unit.

phase difference = [1]

- (ii) Calculate the ratio

$$\frac{\text{intensity of wave R}}{\text{intensity of wave S}}.$$

ratio = [2]

21 (a) State what is meant by the *frequency* of a progressive wave.

9702/21/O/N/16/Q4

.....

[2]

- (b) A cathode-ray oscilloscope (c.r.o.) is used to determine the frequency of the sound emitted by a loudspeaker. The trace produced on the screen of the c.r.o. is shown in Fig. 4.1.

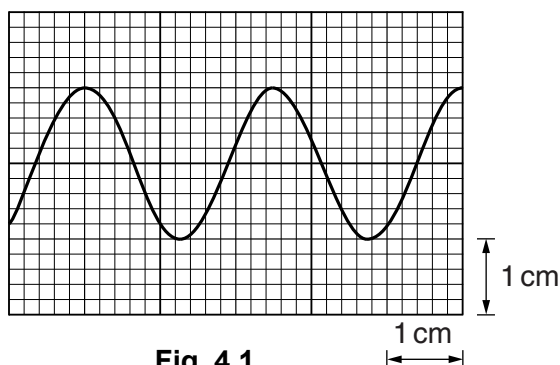


Fig. 4.1

The time-base setting of the c.r.o. is $250 \mu\text{s cm}^{-1}$.

Show that the frequency of the sound wave is 1600 Hz.

[2]

- (c) The loudspeaker in (b) emits the sound in all directions. A person attaches the loudspeaker to a string and then swings the loudspeaker at a constant speed in a horizontal circle above his head.

An observer, standing a large distance away from the loudspeaker, hears sound of maximum frequency 1640 Hz. The speed of sound in air is 330 m s^{-1} .

- (i) Determine the speed of the loudspeaker.

speed = m s^{-1} [2]

- (ii) Describe and explain, qualitatively, the variation in the frequency of the sound heard by the observer.

.....

[2]

22 (a) State the conditions required for the formation of stationary waves.
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.....
.....
.....[2]

(b) One end of a string is attached to a vibrator. The string is stretched by passing the other end over a pulley and attaching a load, as illustrated in Fig. 4.1.

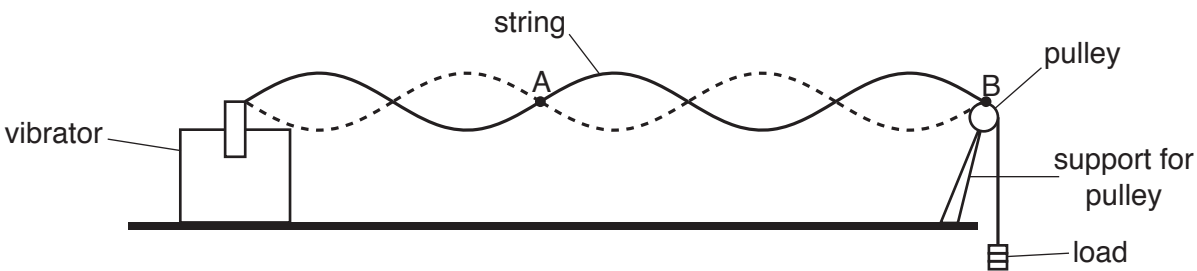


Fig. 4.1

The frequency of vibration of the vibrator is adjusted to 250 Hz and a transverse wave travels along the string with a speed of 12 m s⁻¹. The wave is reflected at the pulley and a stationary wave forms on the string.

Fig. 4.2 shows the string between points A and B at time $t = t_1$.

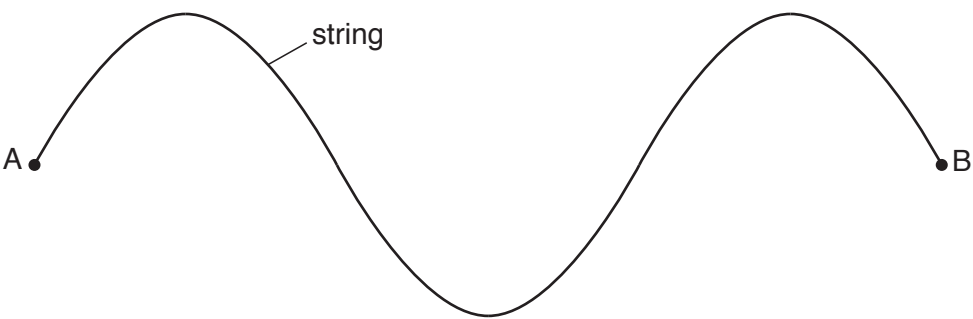


Fig. 4.2

At time $t = t_1$ the string has maximum displacement.

(i) Calculate the distance AB.

distance =m [2]

(ii) On Fig. 4.2, sketch the position of the string between A and B at times

1. $t = t_1 + 2.0 \text{ ms}$ (label this line P),
2. $t = t_1 + 5.0 \text{ ms}$ (label this line Q).

[3]

9702/21/M/J/017/Q5

23 (a) Describe the Doppler effect.

.....

[1]

- (b) A car travels with a constant velocity along a straight road. The car horn with a frequency of 400 Hz is sounded continuously. A stationary observer on the roadside hears the sound from the horn at a frequency of 360 Hz.
 The speed of sound is 340 m s^{-1} .

Determine the magnitude v , and the direction, of the velocity of the car relative to the observer.

$v = \dots\dots\dots \text{m s}^{-1}$

direction

[3]

24 (a) Define the *frequency* of a sound wave.

9702/22/M/J/017/Q5

.....[1]

(b) A sound wave travels through air. Describe the motion of the air particles relative to the direction of travel of the sound wave.

.....[1]

(c) The sound wave emitted from the horn of a stationary car is detected with a microphone and displayed on a cathode-ray oscilloscope (c.r.o.), as shown in Fig. 5.1.

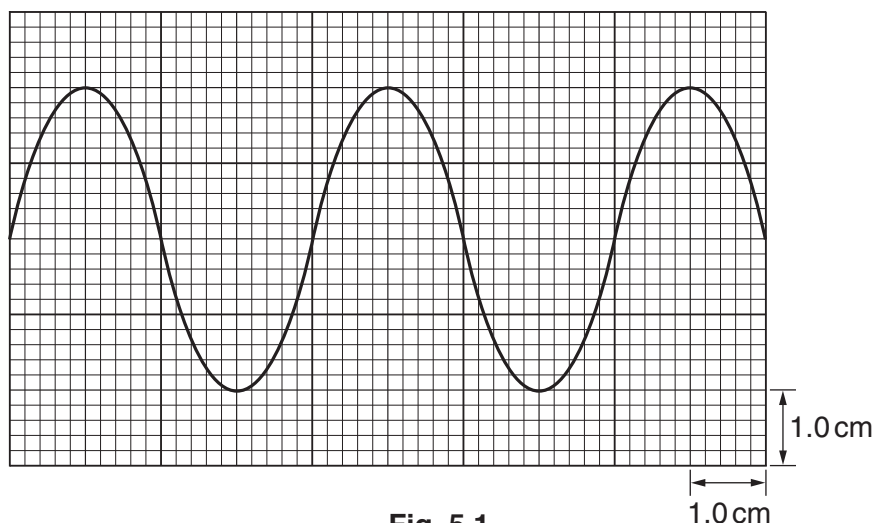


Fig. 5.1

The y -axis setting is 5.0 mV cm^{-1} .

The time-base setting is 0.50 ms cm^{-1} .

(i) Use Fig. 5.1 to determine the frequency of the sound wave.

frequency = Hz [2]

(ii) The horn of the car sounds continuously. Describe the changes to the trace seen on the c.r.o. as the car travels at constant speed

1. directly towards the stationary microphone,

.....

2. directly away from the stationary microphone.

.....[3]

25 (a) State the difference between a stationary wave and a progressive wave in terms of

(i) the energy transfer along the wave,

.....
[1]

(ii) the phase of two adjacent vibrating particles.

.....
[1]

(b) A tube is open at both ends. A loudspeaker, emitting sound of a single frequency, is placed near one end of the tube, as shown in Fig. 3.1.

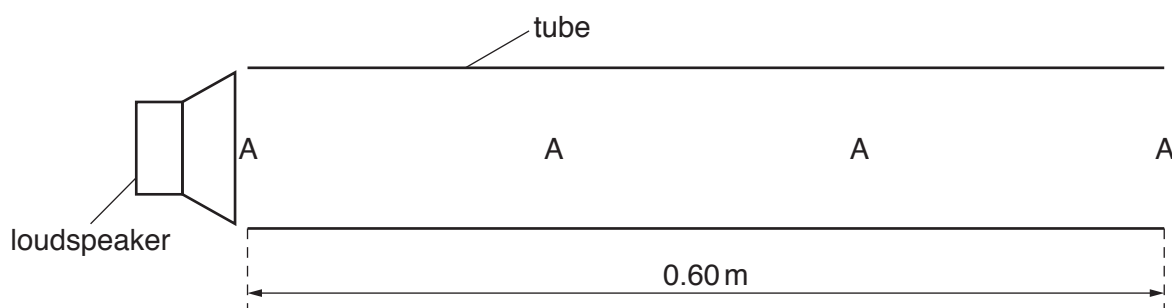


Fig. 3.1

The speed of the sound in the tube is 340 m s^{-1} . The length of the tube is 0.60 m. A stationary wave is formed with an antinode A at each end of the tube and two antinodes inside the tube.

(i) State what is meant by an *antinode* of the stationary wave.

.....
[1]

(ii) State the distance between a node and an adjacent antinode.

distance = m [1]

(iii) Determine, for the sound in the tube,

1. the wavelength,

wavelength = m [1]

2. the frequency.

frequency = Hz [2]

- (iv) Determine the minimum frequency of the sound from the loudspeaker that produces a stationary wave in the tube.

minimum frequency = Hz [2]

26 (a) State the conditions required for the formation of a stationary wave.

9702/22/O/N/17/Q4

.....
.....
.....[2]

- (b) A horizontal string is stretched between two fixed points X and Y. The string is made to vibrate vertically so that a stationary wave is formed. At one instant, each particle of the string is at its maximum displacement, as shown in Fig. 4.1.

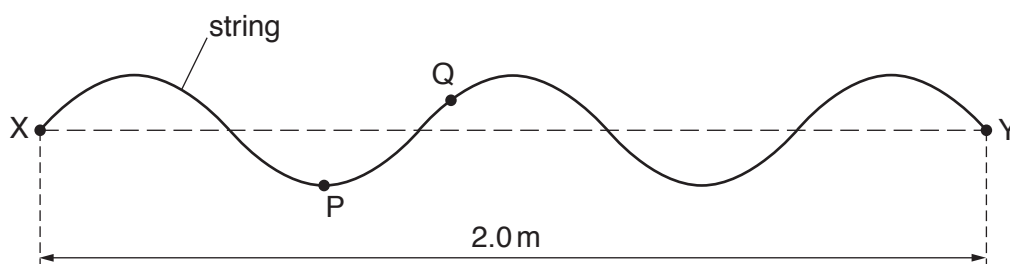


Fig. 4.1

P and Q are two particles of the string. The string vibrates with a frequency of 40 Hz. Distance XY is 2.0 m.

- (i) State the number of antinodes in the stationary wave.

number =[1]

- (ii) Determine the minimum time taken for the particle P to travel from its lowest point to its highest point.

time taken = s [2]

- (iii) State the phase difference, with its unit, between the vibrations of particle P and of particle Q.

phase difference =[1]

- (iv) Determine the speed of a progressive wave along the string.

speed = ms^{-1} [2]

- 27 (a) By reference to the direction of propagation of energy, explain what is meant by a *longitudinal* wave.

.....
[1]

- (b) A car horn emits a sound wave of frequency 800 Hz. A microphone and a cathode-ray oscilloscope (c.r.o.) are used to analyse the sound wave. The waveform displayed on the c.r.o. screen is shown in Fig. 4.1.

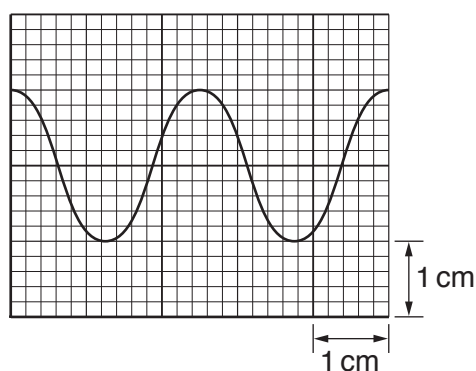


Fig. 4.1

Determine the time-base setting, in s cm^{-1} , of the c.r.o.

time-base setting = s cm^{-1} [3]

- (c) The intensity I of the sound at a distance r from the car horn in (b) is given by the expression

$$I = \frac{k}{r^2}$$

where k is a constant.

Fig. 4.2 shows the car in (b) on a road.

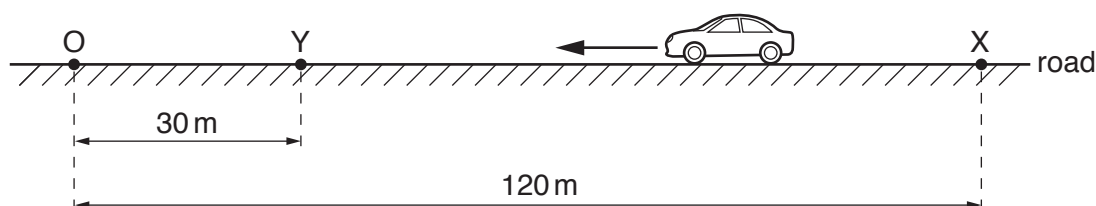


Fig. 4.2

- (c) The intensity I of the sound at a distance r from the car horn in (b) is given by the expression

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where k is a constant.

Fig. 4.2 shows the car in (b) on a road.

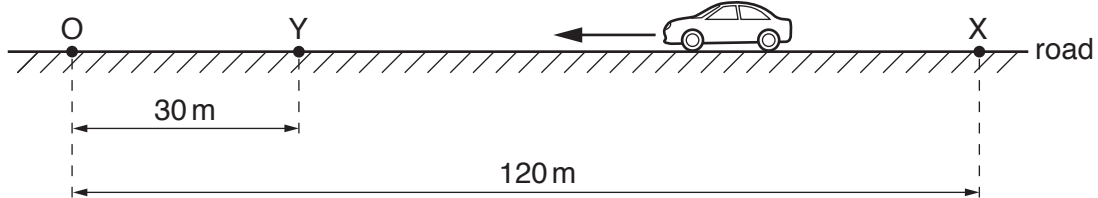


Fig. 4.2

An observer stands at point O. Initially the car is parked at point X which is 120 m away from point O. The car then moves directly towards the observer and stops at point Y, a distance of 30 m away from O.

The car horn continuously emits sound when the car is moving between points X and Y.

- (i) The sound wave at point O has amplitude A_X when the car is at X and has amplitude A_Y when the car is at Y.

Calculate the ratio $\frac{A_Y}{A_X}$.

ratio = [3]

- (ii) When the car is parked at X, the frequency of the sound from the horn that is detected by the observer is 800 Hz. As the car moves from X to Y, the maximum change in the detected frequency is 16 Hz. The speed of the sound in air is 330 m s^{-1} .

Determine, to two significant figures,

1. the minimum wavelength of the sound detected by the observer,

wavelength = m [2]

2. the maximum speed of the car.

speed = m s^{-1} [2]

28 (a) For a progressive wave, state what is meant by

9702/21/M/J/18/Q4

(i) the *period*,

.....
.....[1]

(ii) the *wavelength*.

.....
.....[1]

(b) Fig. 4.1 shows the variation with time t of the displacement x of two progressive waves P and Q passing the same point.

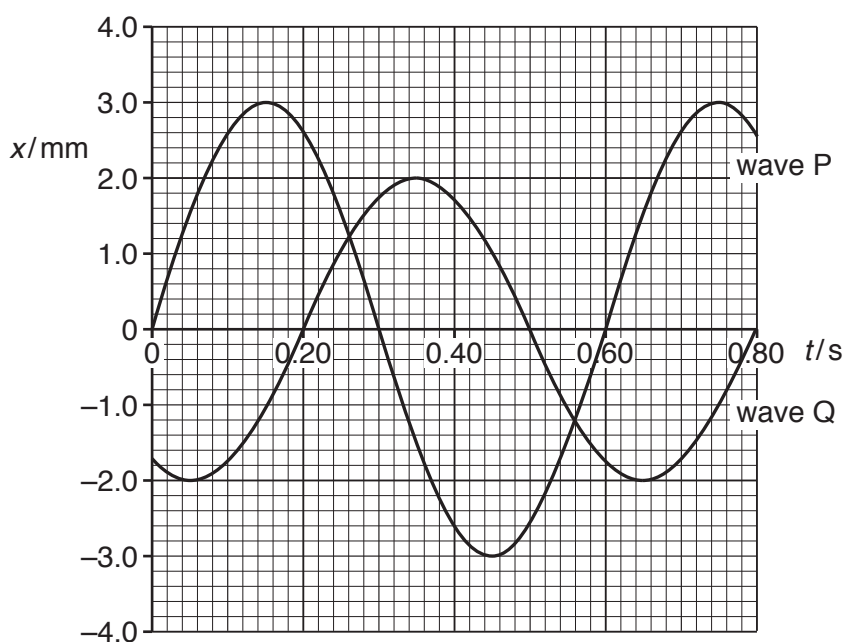


Fig. 4.1

The speed of the waves is 20 cm s^{-1} .

(i) Calculate the wavelength of the waves.

wavelength = cm [2]

(ii) Determine the phase difference between the two waves.

(iii) Calculate the ratio $\frac{\text{intensity of wave Q}}{\text{intensity of wave P}}$.
phase difference = ° [1]

ratio = [2]

(iv) The two waves superpose as they pass the same point. Use Fig. 4.1 to determine the resultant displacement at time $t = 0.45\text{ s}$.

displacement = mm [1]

- 29 (a) (i) Define the *wavelength* of a progressive wave.

.....
[1]

- (ii) State what is meant by an *antinode* of a stationary wave.

.....
[1]

- (b) A loudspeaker producing sound of constant frequency is placed near the open end of a pipe, as shown in Fig. 4.1.

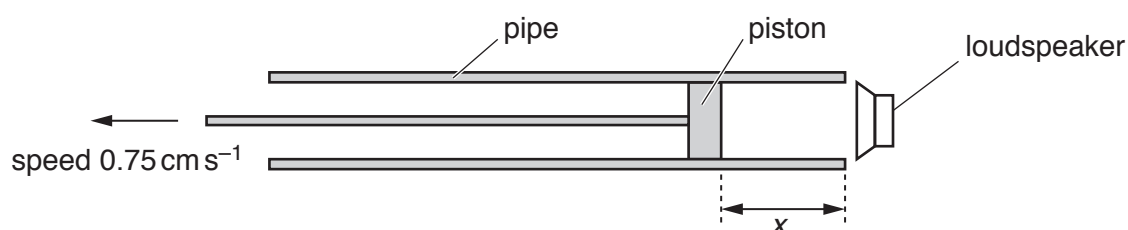


Fig. 4.1

A movable piston is at distance x from the open end of the pipe. Distance x is increased from $x = 0$ by moving the piston to the left with a constant speed of 0.75 cm s^{-1} .

The speed of the sound in the pipe is 340 m s^{-1} .

- (i) A much louder sound is first heard when $x = 4.5 \text{ cm}$. Assume that there is an antinode of a stationary wave at the open end of the pipe.

Determine the frequency of the sound in the pipe.

frequency = Hz [3]

- (ii) After a time interval, a second much louder sound is heard. Calculate the time interval between the first louder sound and the second louder sound being heard.

time interval = s [2]

- 30 (a) Sound waves are longitudinal waves. By reference to the direction of propagation of energy, state what is meant by a *longitudinal wave*.

.....
 [1]

- (b) A stationary sound wave in air has amplitude A . In an experiment, a detector is used to determine A^2 . The variation of A^2 with distance x along the wave is shown in Fig. 4.1.

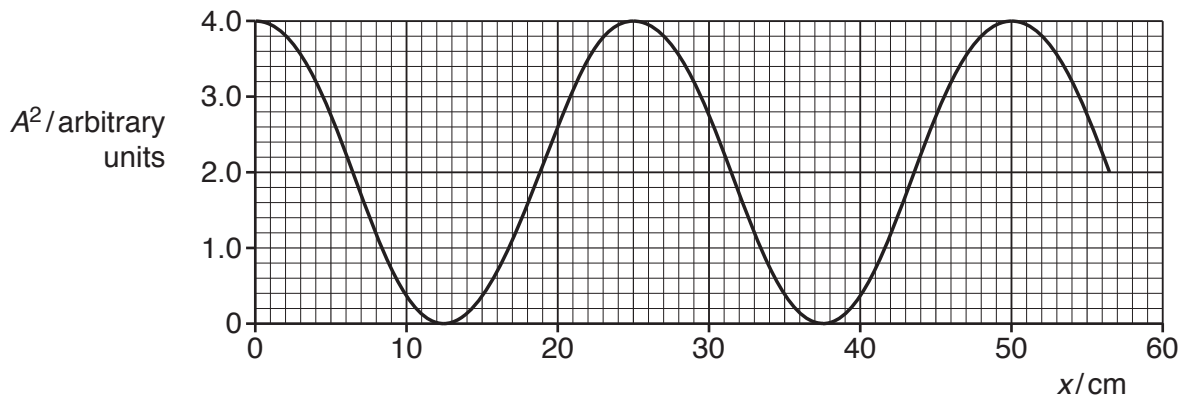


Fig. 4.1

- (i) State the phase difference between the vibrations of an air particle at $x = 25$ cm and the vibrations of an air particle at $x = 50$ cm.

phase difference = ° [1]

- (ii) The speed of the sound in the air is 330 ms^{-1} . Determine the frequency of the sound wave.

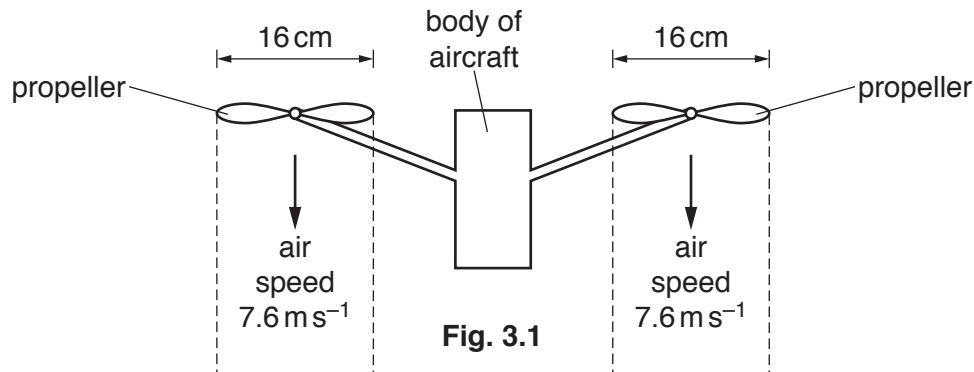
frequency = Hz [3]

- (iii) Determine the ratio

$$\frac{\text{amplitude } A \text{ of wave at } x = 20 \text{ cm}}{\text{amplitude } A \text{ of wave at } x = 25 \text{ cm}}$$

ratio = [2]

- 31 A small remote-controlled model aircraft has two propellers, each of diameter 16 cm. Fig. 3.1 is a side view of the aircraft when hovering.



Air is propelled vertically downwards by each propeller so that the aircraft hovers at a fixed position. The density of the air is 1.2 kg m^{-3} . Assume that the air from each propeller moves with a constant speed of 7.6 m s^{-1} in a uniform cylinder of diameter 16 cm. Also assume that the air above each propeller is stationary.

- (a) Show that, in a time interval of 3.0 s, the mass of air propelled downwards by **one** propeller is 0.55 kg.

[3]

- (b) Calculate:

- (i) the increase in momentum of the mass of air in (a)

increase in momentum = N s [1]

- (ii) the downward force exerted on this mass of air by the propeller.

force = N [1]

(c) State:

- (i) the upward force acting on **one** propeller

force = N [1]

- (ii) the name of the law that explains the relationship between the force in **(b)(ii)** and the force in **(c)(i)**.

..... [1]

(d) Determine the mass of the aircraft.

mass = kg [1]

- (e) In order for the aircraft to hover at a very high altitude (height), the propellers must propel the air downwards with a greater speed than when the aircraft hovers at a low altitude. Suggest the reason for this.

.....
..... [1]

- (f) When the aircraft is hovering at a high altitude, an electric fault causes the propellers to stop rotating. The aircraft falls vertically downwards. When the aircraft reaches a constant speed of 22 m s^{-1} , it emits sound of frequency 3.0 kHz from an alarm. The speed of the sound in the air is 340 m s^{-1} .

Determine the frequency of the sound heard by a person standing vertically below the falling aircraft.

frequency = Hz [2]

- 32 (a) State what is meant by the *wavelength* of a progressive wave.

.....
 [1]

- (b) A cathode-ray oscilloscope (CRO) is used to analyse a sound wave. The screen of the CRO is shown in Fig. 5.1.

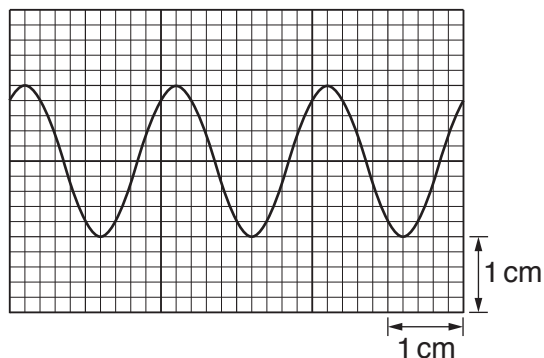


Fig. 5.1

The time-base setting of the CRO is 2.5 ms cm^{-1} .

Determine the frequency of the sound wave.

frequency = Hz [2]

- (c) The source emitting the sound in (b) is at point A. Waves travel from the source to point C along two different paths, AC and ABC, as shown in Fig. 5.2.

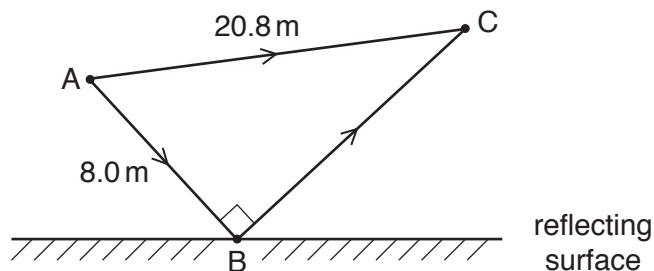


Fig. 5.2 (not to scale)

Distance AB is 8.0 m and distance AC is 20.8 m. Angle ABC is 90° . Assume that there is no phase change of the sound wave due to the reflection at point B. The wavelength of the waves is 1.6 m.

- (i) Show that the waves meeting at C have a path difference of 6.4 m.

[1]

- (ii) Explain why an intensity maximum is detected at point C.

.....

 [2]

- (iii) Determine the difference between the times taken for the sound to travel from the source to point C along the two different paths.

time difference = s [2]

- (iv) The wavelength of the sound is gradually increased. Calculate the wavelength of the sound when an intensity maximum is next detected at point C.

wavelength = m [1]

- 33 Two progressive sound waves Y and Z meet at a fixed point P. The variation with time t of the displacement x of each wave at point P is shown in Fig. 4.1.

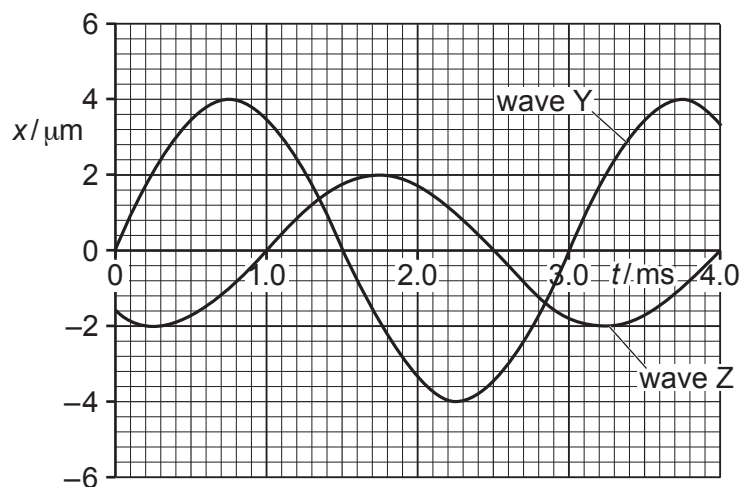


Fig. 4.1

- (a) Use Fig. 4.1 to state **one** quantity of waves Y and Z that is:

(i) the same

..... [1]

(ii) different.

..... [1]

- (b) State and explain whether waves Y and Z are coherent.

..... [1]

- (c) Determine the phase difference between the waves.

phase difference = ° [1]

- (d) The two waves superpose at P. Use Fig. 4.1 to determine the resultant displacement at time $t = 0.75$ ms.

resultant displacement = μm [1]

- (e) The intensity of wave Y at point P is I .

Determine, in terms of I , the intensity of wave Z.

intensity = [2]

- (f) The speed of wave Z is 330 ms^{-1} .

Determine the wavelength of wave Z.

wavelength = m [3]

34. 9702/21/O/N/20/Q5

A progressive wave Y passes a point P. The variation with time t of the displacement x for the wave at P is shown in Fig. 5.1.

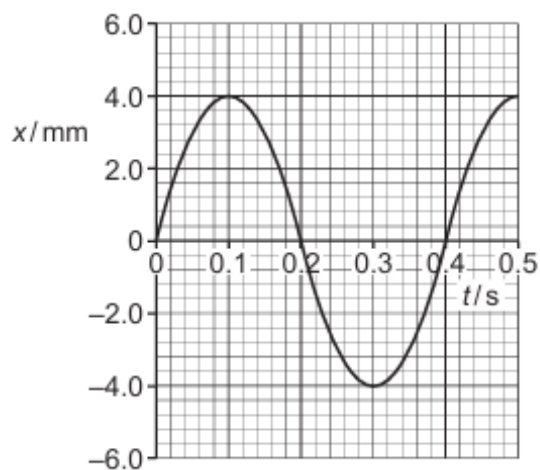


Fig. 5.1

The wave has a wavelength of 8.0 cm.

- (a) Determine the speed of the wave.

speed = ms^{-1} [2]

- (b) A second wave Z has wavelength 8.0 cm and amplitude 2.0 mm at point P. Waves Y and Z have the same speed.

For the waves at point P, calculate the ratio

$$\frac{\text{intensity of wave Z}}{\text{intensity of wave Y}}.$$

ratio = [3]

[Total: 5]

35. 9702/22/F/M/21/Q5

A source of sound is attached to a rope and then swung at a constant speed in a horizontal circle, as illustrated in Fig. 5.1.

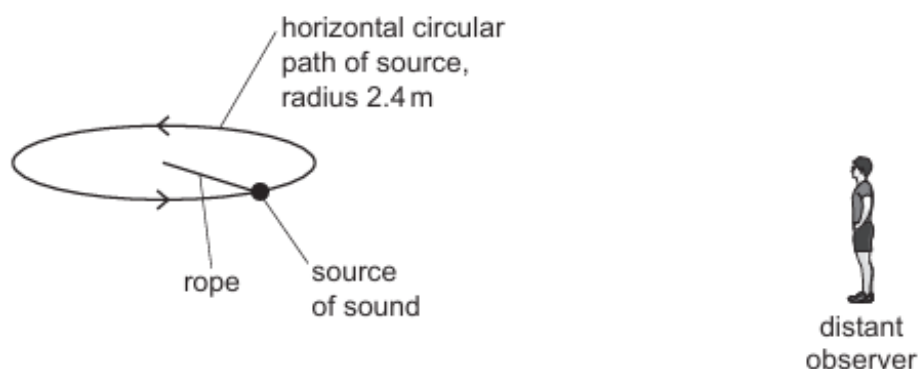


Fig. 5.1 (not to scale)

The source moves with a speed of 12.0 m s^{-1} and emits sound of frequency 951 Hz . The speed of the sound in the air is 330 m s^{-1} . An observer, standing a very long distance away from the source, hears the sound.

- (a) Calculate the minimum frequency, to three significant figures, of the sound heard by the observer.

minimum frequency = Hz [2]

- (b) The circular path of the source has a radius of 2.4 m .

Determine the shortest time interval between the observer hearing sound of minimum frequency and the observer hearing sound of maximum frequency.

time interval = s [2]

[Total: 4]

36. 9702/21/O/N/21/Q4

- (a) By reference to the direction of transfer of energy, state what is meant by a *longitudinal wave*.

.....
 [1]

- (b) A vehicle travels at constant speed around a wide circular track. It continuously sounds its horn, which emits a single note of frequency 1.2 kHz. An observer is a large distance away from the track, as shown in the view from above in Fig. 4.1.

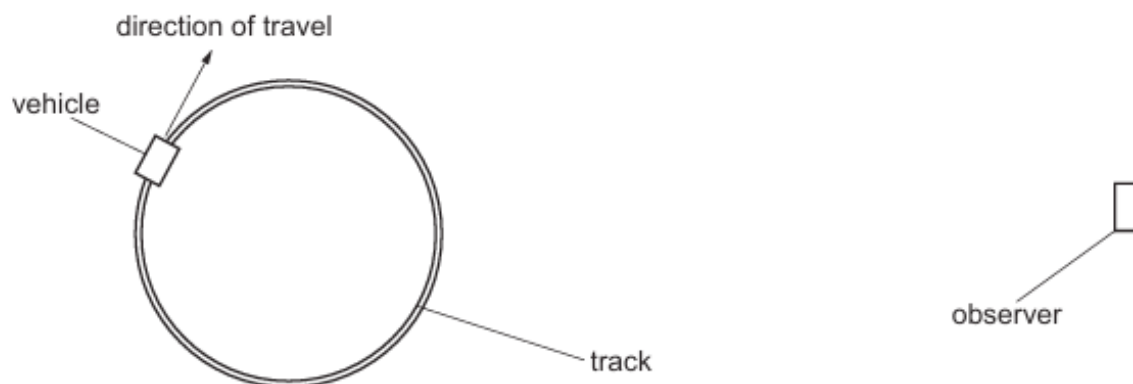


Fig. 4.1 (not to scale)

Fig. 4.2 shows the variation with time of the frequency f of the sound of the horn that is detected by the observer. The time taken for the vehicle to travel once around the track is T .

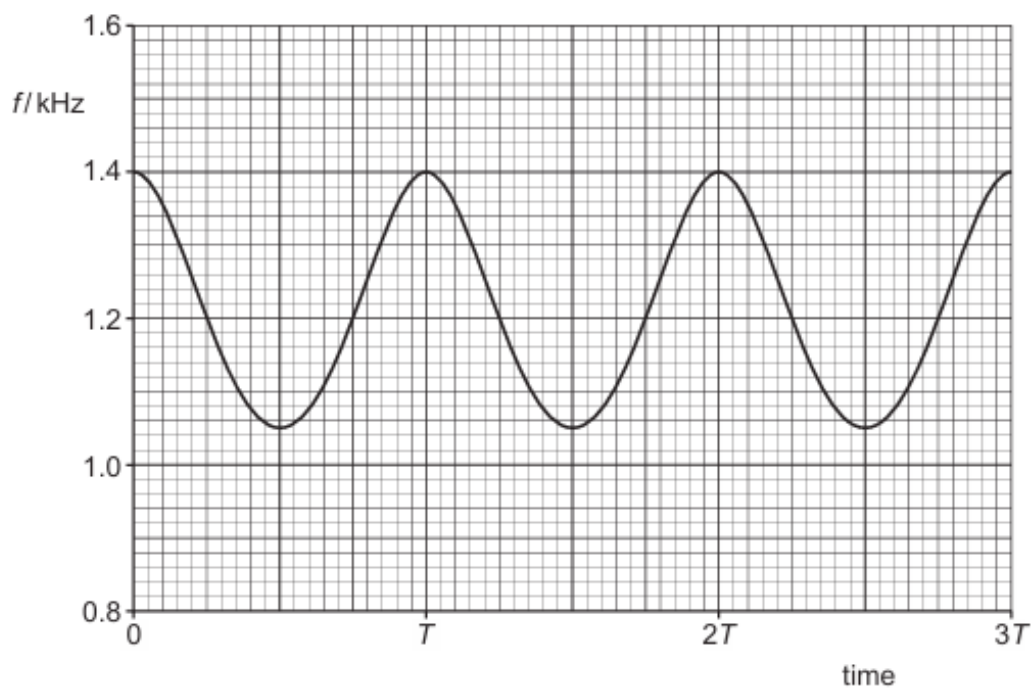


Fig. 4.2

- (i) Explain why the frequency of the sound detected by the observer is sometimes above and sometimes below 1.2 kHz.

.....
.....
.....
..... [2]

- (ii) State the name of the phenomenon in (b)(i).

..... [1]

- (iii) On Fig. 4.1, mark with a letter X the position of the vehicle when it emitted the sound that is detected at time T . [1]

- (iv) On Fig. 4.1, mark with a letter Y the position of the vehicle when it emitted the sound that is detected at time $\frac{9T}{4}$. [1]

- (c) The speed of the sound in the air is 320 m s^{-1} .

Use Fig. 4.2 to determine the speed of the vehicle in (b).

speed = m s^{-1} [3]

[Total: 9]

37. 9702/22/O/N/21/Q4

A child sits on the ground next to a remote-controlled toy car. At time $t = 0$, the car begins to move in a straight line directly away from the child. The variation with time t of the velocity of the car along this line is shown in Fig. 4.1.

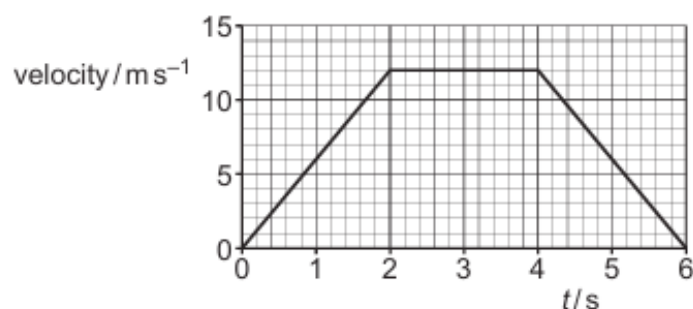


Fig. 4.1

The car's horn continually emits sound of frequency 925 Hz between time $t = 0$ and time $t = 6.0$ s. The speed of the sound in the air is 338 m s^{-1} .

- (a) Describe qualitatively the variation, if any, in the frequency of the sound heard, by the child, that was emitted from the car horn:

- (i) from time $t = 0$ to time $t = 2.0$ s

..... [1]

- (ii) from time $t = 4.0$ s to time $t = 6.0$ s.

..... [1]

- (b) Determine the frequency, to three significant figures, of the sound heard, by the child, that was emitted from the car horn at time $t = 3.0$ s.

frequency = Hz [2]

- (c) Determine the time taken for the sound emitted at time $t = 4.0$ s to travel to the child.

time taken = s [2]

[Total: 6]

38. 9702/22/F/M/22/Q5

- (a) (i) State the conditions required for the formation of a stationary wave.

.....
.....
.....
..... [2]

- (ii) State the phase difference between any two vibrating particles in a stationary wave between two adjacent nodes.

phase difference =° [1]

- (b) A motorcycle is travelling at 13.0 ms^{-1} along a straight road. The rider of the motorcycle sees a pedestrian standing in the road directly ahead and operates a horn to emit a warning sound. The pedestrian hears the warning sound from the horn at a frequency of 543 Hz . The speed of the sound in the air is 334 ms^{-1} .

- (i) Calculate the frequency, to three significant figures, of the sound emitted by the horn.

frequency = Hz [2]

- (ii) The motorcycle rider passes the stationary pedestrian and then moves directly away from her. As the rider moves away, he operates the horn for a second time. The pedestrian now hears sound that is increasing in frequency.

State the variation, if any, in the speed of the motorcycle when the rider operates the horn for the second time.

..... [1]

- (c) A beam of vertically polarised monochromatic light is incident normally on a polarising filter, as shown in Fig. 5.1.

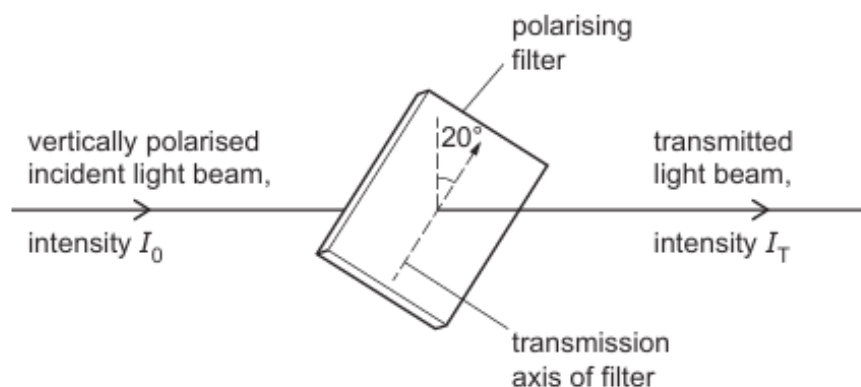


Fig. 5.1

The filter is positioned with its transmission axis at an angle of 20° to the vertical. The incident light has intensity I_0 and the transmitted light has intensity I_T .

- (i) By considering the ratio $\frac{I_T}{I_0}$, calculate the ratio

$$\frac{\text{amplitude of transmitted light}}{\text{amplitude of incident light}}.$$

Show your working.

ratio = [3]

- (ii) The filter is now rotated, about the direction of the light beam, from its starting position shown in Fig. 5.1. The direction of rotation is such that the angle of the transmission axis to the vertical initially increases.

Calculate the minimum angle through which the filter must be rotated so that the intensity of the transmitted light returns to the value that it had when the filter was at its starting position.

angle = $^\circ$ [1]

39. 9702/21/O/N/22/Q4

(a) Polarisation is a phenomenon associated with light waves but not with sound waves.

(i) State the meaning of polarisation.

.....

.....

..... [1]

(ii) State why light waves can be plane polarised but sound waves cannot.

.....

.....

..... [1]

(b) Two polarising filters A and B are positioned so that their planes are parallel to each other and perpendicular to a central axis line XY, as shown in Fig. 4.1.

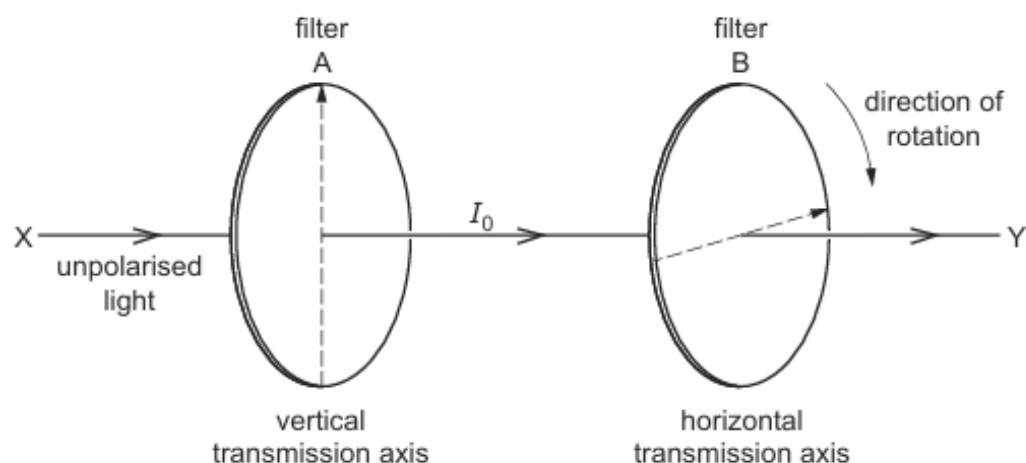


Fig. 4.1

The transmission axis of filter A is vertical and the transmission axis of filter B is horizontal.

Unpolarised light of a single frequency is directed along the line XY from a source positioned at X. The light emerging from filter A is vertically plane polarised and has intensity I_0 .

Filter B is rotated from its starting position about the line XY, as shown in Fig. 4.1.

After rotation, the intensity of the light emerging from filter B is $\frac{1}{4} I_0$.

Calculate the angle of rotation of filter B from its starting position.

angle of rotation = ° [3]

- (c) A microwave of intensity I_0 and amplitude A_0 meets another microwave of the same frequency and of intensity $\frac{1}{4} I_0$ travelling in the opposite direction. Both microwaves are vertically plane polarised and superpose where they meet.

- (i) Explain, without calculation, why these two waves cannot form a stationary wave with zero amplitude at its nodes.

.....
.....
..... [2]

- (ii) Determine, in terms of A_0 , the maximum amplitude of the wave formed.

maximum amplitude = A_0 [3]

[Total: 10]

40. 9702/22/O/N/22/Q5

- (a) Two progressive sound waves meet to form a stationary wave. The two waves have the same amplitude, wavelength, frequency and speed.

State the other condition that must be fulfilled by the two waves in order for them to produce the stationary wave.

..... [1]

- (b) A stationary wave is formed on a string that is stretched between two fixed points A and B. Fig. 5.1 shows the string at time $t = 0$ when each point is at its maximum displacement.



Fig. 5.1

Distance AB is 0.80 m. The period of the stationary wave is 0.016 s.

- (i) On Fig. 5.1, sketch a solid line to show the position of the string:

- at time $t = 0.004$ s (label this line P)
- at time $t = 0.024$ s (label this line Q).

[2]

- (ii) Determine the speed of a progressive wave along the string.

speed = ms^{-1} [3]

- (c) A beam of vertically polarised light of intensity I_0 is incident normally on a polarising filter that has its transmission axis at 30° to the vertical, as shown in Fig. 5.2.

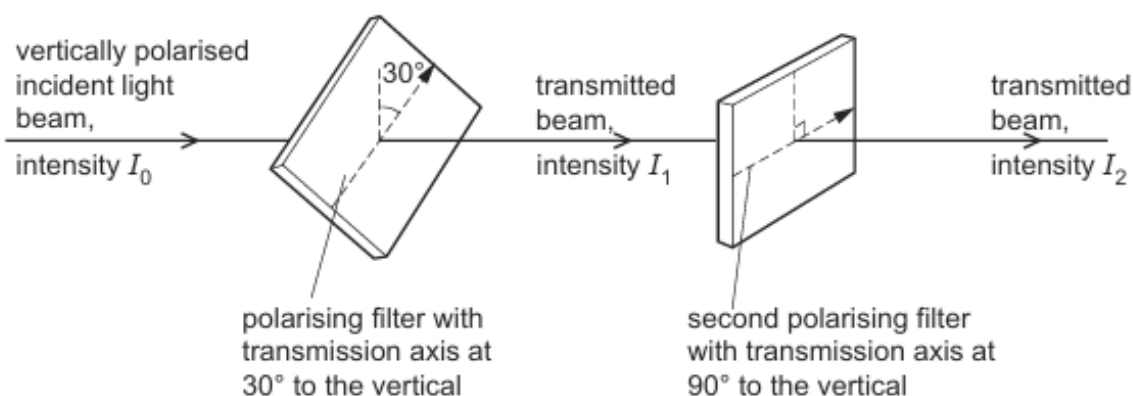


Fig. 5.2

The transmitted light from the first polarising filter has intensity I_1 . This light is then incident normally on a second polarising filter that has its transmission axis at 90° to the vertical. The transmitted light from the second filter has intensity I_2 .

Calculate:

- (i) the ratio $\frac{I_1}{I_0}$

$$\frac{I_1}{I_0} = \dots\dots\dots [2]$$

- (ii) the ratio $\frac{I_2}{I_0}$.

$$\frac{I_2}{I_0} = \dots\dots\dots [2]$$

[Total: 10]

41. 9702/23/O/N/22/Q4

- (a) A progressive longitudinal wave travels through a medium from left to right. Fig. 4.1 shows the positions of some of the particles of the medium at time t_0 and a graph showing the particle displacements at the same time t_0 .

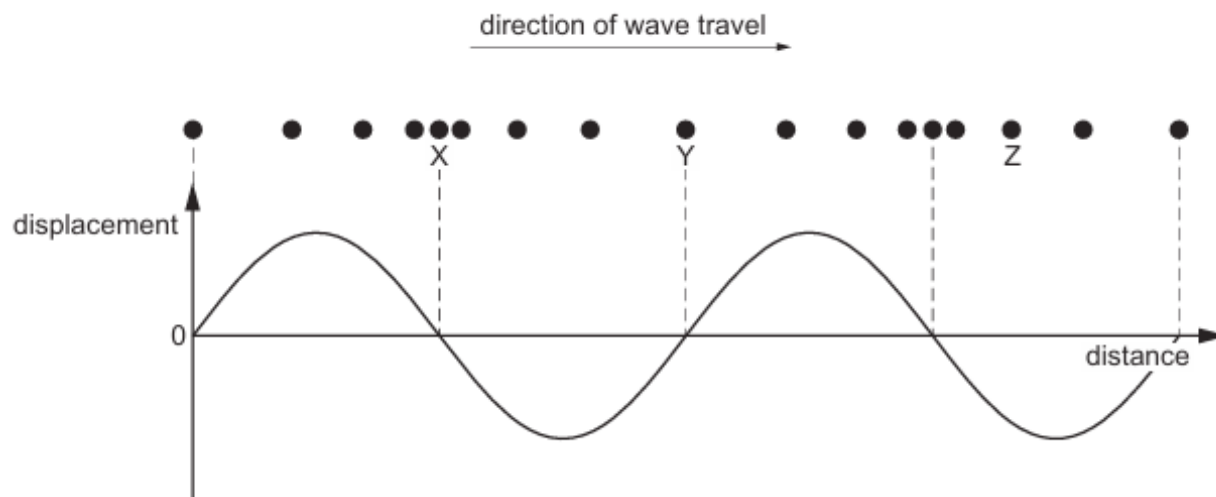


Fig. 4.1

Particle displacements to the right of their equilibrium positions are shown as positive on the graph and particle displacements to the left are shown as negative on the graph.

The period of the wave is T .

- (i) On Fig. 4.1, draw circles around two particles which are exactly one wavelength apart. [1]
- (ii) On Fig. 4.1, sketch a line on the graph to represent the displacements of the particles for the longitudinal wave at time $t_0 + \frac{T}{4}$. [3]
- (iii) State the direction of motion of particle Z at time $t_0 + \frac{T}{4}$. [1]

..... [1]

- (b) The frequency of the wave in (a) is 16 kHz. The distance between particles X and Y is 0.19 m.

Calculate the speed of the wave as it travels through the medium.

speed = ms^{-1} [3]

- (c) A longitudinal sound wave is travelling through a solid. The initial intensity of the wave is I_0 . The frequency of the wave remains constant and the amplitude falls to half of its original value.

Determine, in terms of I_0 , the final intensity of the wave.

intensity = I_0 [2]

- (d) The sound wave in (c) now meets another sound wave travelling in the opposite direction.

- (i) State a condition necessary for these two waves to form a stationary wave.

..... [1]

- (ii) State **two** ways in which a stationary wave differs from a progressive wave.

1

.....

2

.....

[2]

[Total: 13]

42. 9702/22/M/J/23/Q5

- (a) A progressive wave travels through a medium. The wave causes a particle of the medium to vibrate along a line P. The energy of the wave propagates along a line Q.

Compare the directions of lines P and Q if the wave is:

- (i) a transverse wave

..... [1]

- (ii) a longitudinal wave.

..... [1]

- (b) A tube is closed at one end. A loudspeaker is placed near the other end of the tube, as shown in Fig. 5.1.

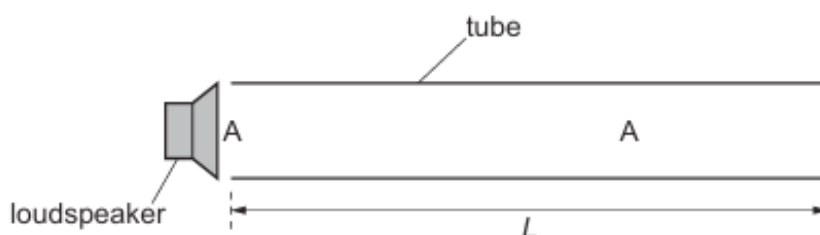


Fig. 5.1 (not to scale)

The loudspeaker emits sound of frequency 1.7 kHz . The speed of sound in the air in the tube is 340 m s^{-1} . A stationary wave is formed with an antinode A at the open end of the tube. There is only one other antinode A inside the tube, as shown in Fig. 5.1.

Determine:

- (i) the wavelength of the sound

wavelength = m [2]

- (ii) the length L of the tube

$L =$ m [1]

- (iii) the maximum wavelength of the sound from the loudspeaker that can produce a stationary wave in the tube.

maximum wavelength =m [1]

- (c) Two polarising filters are arranged so that their planes are vertical and parallel. The first filter has its transmission axis at an angle of 35° to the vertical and the second filter has its transmission axis at angle α to the vertical, as shown in Fig. 5.2.

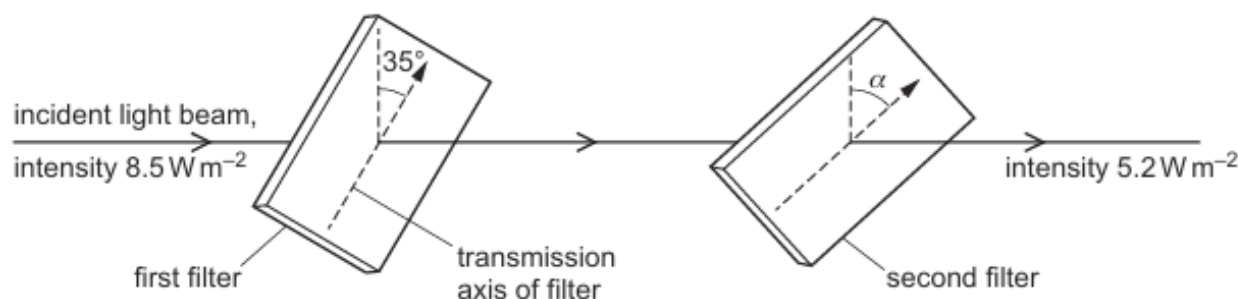


Fig. 5.2

Angle α is greater than 35° and less than 90° . A beam of vertically polarised light of intensity 8.5 W m^{-2} is incident normally on the first filter.

- (i) Show that the intensity of the light transmitted by the first filter is 5.7 W m^{-2} .

[1]

- (ii) The intensity of the light transmitted by the second filter is 5.2 W m^{-2} .

Calculate angle α .

$\alpha = \dots\dots\dots^\circ$ [2]

[Total: 9]

43. 9702/21/O/N/23/Q5

A train travels at a constant high speed along a straight horizontal track towards an observer standing adjacent to the track, as shown in Fig. 5.1.

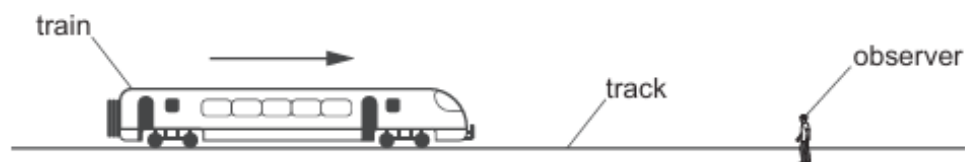


Fig. 5.1

The train sounds its horn continuously as it approaches the observer, from time $t = 0$ until it is well past the observer at time $t = t_2$. The train passes the observer at time $t = t_1$. The horn emits a sound wave of constant frequency f_s .

- (a) On Fig. 5.2, sketch the variation of the frequency of sound heard by the observer with time t , from time $t = 0$ to $t = t_2$.

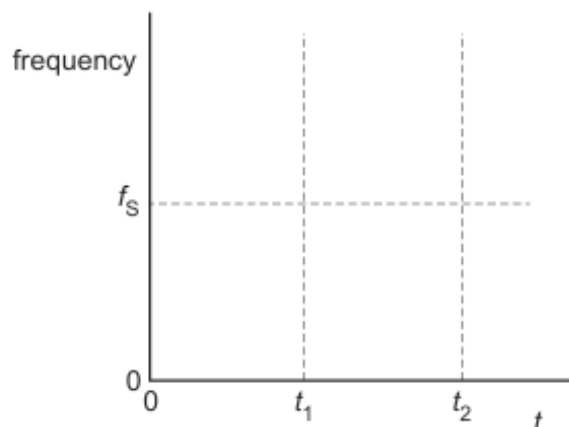


Fig. 5.2

[1]

- (b) At a particular time, the sound waves at the observer have an intensity of $4.7 \times 10^{-3} \text{ W m}^{-2}$. The waves at the observer are incident at right angles on a circular detector of radius 2.8 cm.

Calculate the power P of the waves incident on the detector.

$P = \dots\dots\dots \text{ W}$ [3]

[Total: 4]

44. 9702/23/O/N/23/Q6

A train travels at constant speed along a straight horizontal track towards an observer standing adjacent to the track, as shown in Fig. 6.1.

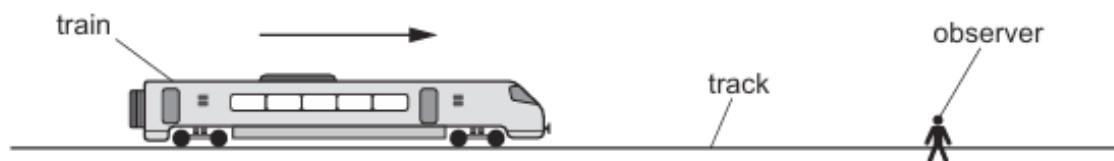


Fig. 6.1

The train sounds its horn continuously as it approaches the observer. The horn emits a sound of constant frequency 251 Hz. The frequency of sound heard by the observer is 291 Hz. The speed of sound in air is 340 m s^{-1} .

(a) Calculate the speed of the train.

speed = m s^{-1} [2]